JAVA CLASS PROGRAMS

1. Student.java

```java
class student {
    int studentId;
    String name;
    int age;

    void display() {
        System.out.println("Student ID: " + studentId);
        System.out.println("Name: " + name);
        System.out.println("Age: " + age);
        System.out.println("----------------");
    }

    public static void main(String[] args) {
        student s1 = new student();
        s1.studentId = 101;
        s1.name = "Rishik";
        s1.age = 18;

        student s2 = new student();
        s2.studentId = 102;
        s2.name = "Lokesh";
        s2.age = 19;

        student s3 = new student();
        s3.studentId = 103;
        s3.name = "Arun Tej";
        s3.age = 20;

        s1.display();
        s2.display();
        s3.display();
    }
}
```

Output:

```
Student ID: 101
Name: Rishik
Age: 19
----------------
Student ID: 102
Name: Lokesh
Age: 18
----------------
Student ID: 103
Name: Arun Tej
Age: 20
----------------
```

2. Employee.java

```java
class employee {
    int employeeId;
    String name;
    double salary;

    employee(int id, String n, double s) {
        employeeId = id;
        name = n;
        salary = s;
    }

    void display() {
        System.out.println("ID: " + employeeId);
        System.out.println("Name: " + name);
        System.out.println("Salary: " + salary);
    }

    public static void main(String[] args) {
        employee e = new employee(101, "Rishik", 25000);
        e.display();
    }
}
```

Output:

```
ID: 101
Name: Rishik
Salary: 25000.0
```

3. BankAccount.java

```
class bankaccount {
    int accountnumber;
    double balance;

    bankaccount(int acc, double bal) {
        accountnumber = acc;
        balance = bal;
    }

    void deposit(double amount) {
        balance += amount;
        System.out.println("Deposited: " + amount);
    }

    void withdraw(double amount) {
        if (amount > 0 && amount <= balance) {
            balance -= amount;
            System.out.println("Withdrawn: " + amount);
        } else {
            System.out.println("Insufficient balance or invalid
amount");
        }
    }

    public static void main(String[] args) {
        bankaccount b = new bankaccount(12345, 5000);

        b.deposit(2000);
        b.withdraw(1000);

        System.out.println("Balance: " + b.balance);
    }
}
```

Output:

```
Deposited: 2000.0
Withdrawn: 1000.0
Balance: 6000.0
```

4. Rectangle.java

```
class rectangle {
    double length, width;

    rectangle(double l, double w) {
        length = l;
        width = w;
    }

    void calculate() {
        double area = length * width;
        double perimeter = 2 * (length + width);

        System.out.println("Area: " + area);
        System.out.println("Perimeter: " + perimeter);
    }

    public static void main(String[] args) {
        rectangle r = new rectangle(10, 5);
        r.calculate();
    }
}
```

Output:

```
Area: 50.0
Perimeter: 30.0
```

5. Book.java

```
class book {
    String ISBN;
    String title;
    double price;

    book(String i, String t, double p) {
        ISBN = i;
        title = t;
        price = p;
    }

    void display() {
        System.out.println("ISBN: " + ISBN);
        System.out.println("Title: " + title);
```

```
        System.out.println("Price: " + price);
    }

    public static void main(String[] args) {
        book b = new book("123456", "Java Programming", 450);
        b.display();
    }
}
```

Output:

```
ISBN: 123456
Title: Java Programming
Price: 450.0
```